



THE KING'S SCHOOL

2009 Higher School Certificate Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Answer each question in a separate booklet

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \frac{x}{(x+1)^2} dx$ **2**

(b) (i) Express $\frac{2x+9}{(2x-1)(x+2)}$ in partial fractions. **2**

(ii) Find $\int \frac{2x+9}{(2x-1)(x+2)} dx$ **1**

(c) (i) Show that $\cos^3 x \sin^{12} x = \cos x \sin^{12} x - \cos x \sin^{14} x$ **1**

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \cos^3 x \sin^{12} x dx$ **2**

(d) Evaluate $\int_1^3 \frac{dx}{(x+1)\sqrt{x}}$ by using the substitution $x = u^2$ or otherwise. **3**

(e) Use integration by parts to evaluate $\int_1^e \frac{\ln x}{x^2} dx$ **4**

End of Question 1

-
- (a) Let $z = \sqrt{2} + \sqrt{2}i$
- (i) Find $|z|$ and $\arg z$ 2
- (ii) Find z^{12} 2
- (b) Find the square roots of $1 + 2\sqrt{2}i$ 3
- (c) (i) On the same Argand diagram carefully sketch the region where
 $|z - 1| \leq |z - 3|$ and $|z - 2| \leq 1$ hold simultaneously. 3
- (ii) Find the greatest possible values for $|z|$ and $\arg z$ in this region. 2
- (d) Let $P(x) = x^4 - 4Ax^3 + 3$, A real
- By considering $P'(x)$, or otherwise, find the values for A for which $P(x) = 0$ has 4 complex roots. 3

End of Question 2

- (a) The roots of $x^3 + x + 1 = 0$ are α, β, γ .

Find a cubic equation whose roots are:

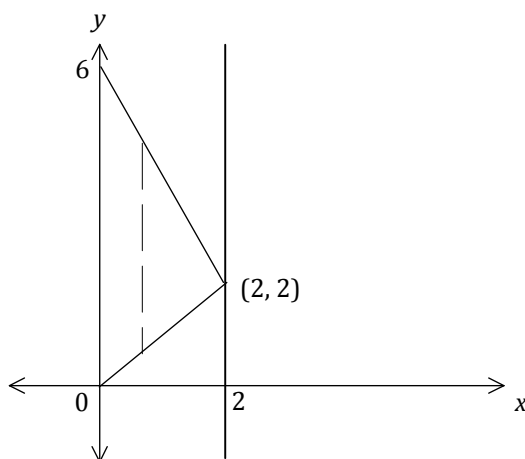
$$\frac{1}{1-\alpha}, \frac{1}{1-\beta}, \frac{1}{1-\gamma}$$

Express your answer in the form $ax^3 + bx^2 + cx + d = 0$

4

- (b) The triangular region bounded by the lines $y = x$, $y = 6 - 2x$ and the y axis is revolved about the line $x = 2$.

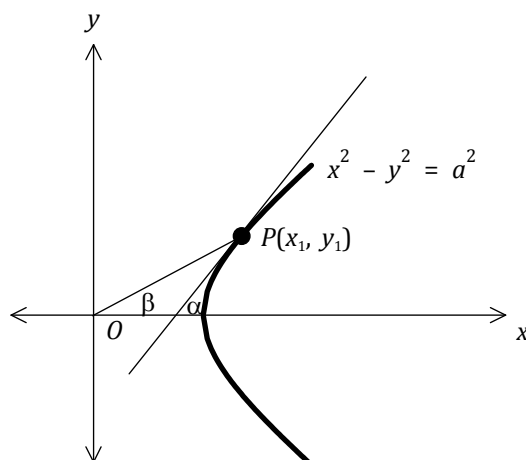
By considering slices of the region parallel to the line $x = 2$, find the volume of the solid of revolution.



5

Question 3 continues on the next page

(c)



The tangent at $P(x_1, y_1)$ in the first quadrant on the hyperbola $x^2 - y^2 = a^2$ meets the x axis at an angle α . The line OP , where O is the origin, meets the x axis at an angle β .

- (i) Prove that the product of the gradients of the line OP and the tangent at P is 1. **3**
- (ii) Deduce that $\alpha + \beta = \frac{\pi}{2}$. **3**

End of Question 3

(a) (i) Show that $1 - x + x^2 - x^3 + \dots + x^{2n} = \frac{1 + x^{2n+1}}{1 + x}$ **1**

(ii) Let $J = \int_0^1 \frac{x^{2n+1}}{1+x} dx$

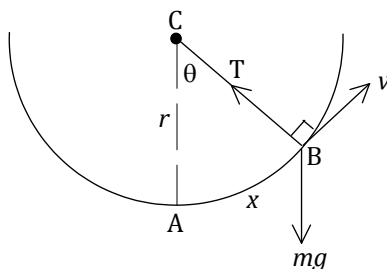
Deduce that $J = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n+1} - \ln 2$ **2**

(iii) Show that $0 < J < \frac{1}{2n+2}$ by considering $\int_0^1 \frac{x^{2n+1}}{1+x} dx$ **2**

(iv) Deduce that $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ **1**

Question 4 continues on the next page

(b)



A simple pendulum consists of a small bob of mass m which is suspended from a fixed point C by a light inextensible string of length r .

The bob is initially at A, vertically below C. Then the bob is displaced through some angle and released from rest.

Suppose at time t the bob is at position B on the circle, as in the diagram.

Let $\angle ACB = \theta$, the arc length AB be x and the linear velocity be $v = \frac{dx}{dt}$

Let T be the tension in the string at time t .

(i) Show that $v = r \frac{d\theta}{dt}$ 2

(ii) By resolving the forces at B in the tangential direction, show that $\frac{dv}{dt} = -g \sin \theta$ 2

(iii) Deduce that $\frac{d^2\theta}{dt^2} = -\frac{g}{r} \sin \theta$ 1

(iv) Suppose the initial angle of release from rest is small.

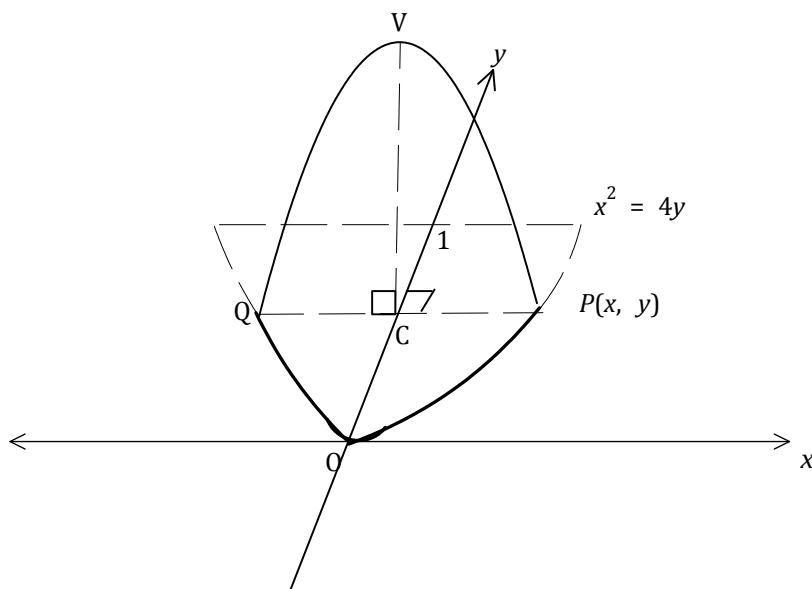
Deduce that the motion of the bob approximates simple harmonic motion and finds its period. 2

(v) If the initial release angle is small, by resolving forces at B in another suitable direction, show that the tension in the string is approximately

$$T = m \left(g + \frac{v^2}{r} \right) \quad 2$$

End of Question 4

(a)



The base of a solid is the region bounded by the parabola $x^2 = 4y$ and the line $y = 1$.

Cross-sections perpendicular to this base and the y axis are parabolic segments with their vertices V directly above the y axis. The diagram shows a typical segment PVQ . All the segments have the property that the vertical height VC is three times the base length PQ .

Let $P(x, y)$ where $x \geq 0$ be a point on the parabola $x^2 = 4y$.

(i) Show that the area of the segment PVQ is $8x^2$. 3

(ii) Find the volume of the solid. 3

Question 5 continues on the next page

- (b) A particle of mass m falls vertically from rest from a point O in a medium whose resistance is mkv , where v is its velocity at any time t , and k is a positive constant.

g is the constant acceleration due to gravity.

Let x be the distance travelled from O by the particle.

- (i) Show that the equation of motion is given by $\ddot{x} = g - kv$ 1

- (ii) Show that the terminal velocity $V = \frac{g}{k}$ 1

- (iii) Use integration to prove that $v = V(1 - e^{-kt})$ 3

- (iv) At the same time as the first particle is released from O another particle of mass m is projected vertically upward from O with initial velocity A .

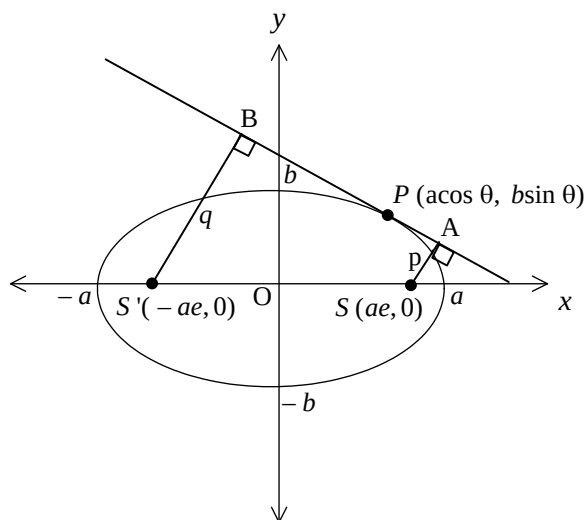
Prove that when this second particle is momentarily at rest the velocity of the first particle is $\frac{AV}{A + V}$ 4

End of Question 5

- (a) (i) Sketch the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$ clearly indicating its foci, directrices and asymptotes. Include on your sketch the points where the hyperbola meets the coordinates axes. 4

- (ii) $P(x_1, y_1)$, $x_1 > 0$, is a point on a branch of the hyperbola. Write down the distance from P to the focus of that branch. 1

(b)



$P(\text{acos}\theta, \text{bsin}\theta)$ is any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$.

$S(ae, 0)$ and $S'(-ae, 0)$ are the foci of the ellipse where e is the eccentricity.

- (i) Prove that the equation of the tangent at $P(\text{acos}\theta, \text{bsin}\theta)$ is $bx \cos\theta + ay \sin\theta - ab = 0$. 3
- (ii) Perpendiculars of lengths p and q are drawn from the foci S and S' to meet the tangent at P at A and B respectively.

Prove that $pq = b^2$. 3

Question 6 continues next page

(iii) Verify that $pq = b^2$ if P is the point $(a, 0)$. 1

(iv) For a particular tangent it is found that $p^2 + q^2 = 6(a^2 - b^2)$ also.

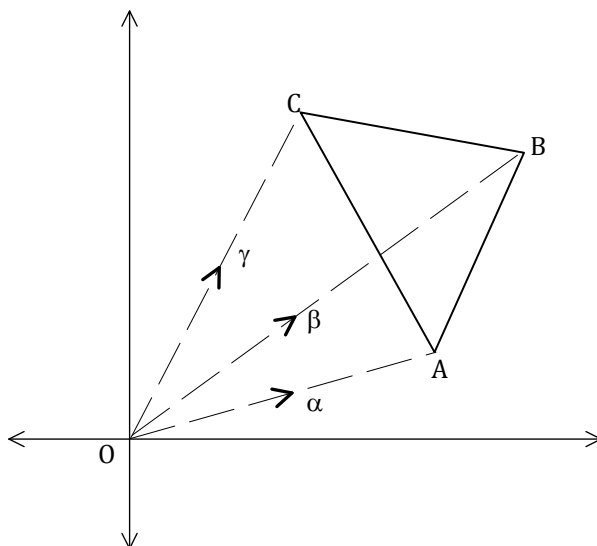
By considering $(p - q)^2$, or otherwise, prove that the ellipse must have an eccentricity $e \geq \frac{1}{2}$. 3

End of Question 6

(a) Let $w = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

(i) Show that $w^3 = 1$ and $1 + w + w^2 = 0$

3



The points A, B, C in the Argand diagram represent the complex numbers α , β , γ , respectively.

ΔABC is equilateral.

(ii) Show that $\alpha - \gamma = w(\gamma - \beta)$

2

(iii) Deduce that $\alpha + w\beta + w^2\gamma = 0$

1

(iv) Explain why α , $w\beta$ and $w^2\gamma$ are the roots of a cubic equation $z^3 + pz + q = 0$.

1

(v) Deduce that $q = -\alpha\beta\gamma$

1

(vi) Prove that $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$

2

Question 7 continues on the next page

(b) Let $u_n = \int_0^{\frac{\pi}{2}} \frac{\sin 2n\theta}{\sin \theta} d\theta$, $n = 1, 2, 3, \dots$

(i) Use the trigonometric relationship

$$\sin 2n\theta - \sin 2(n-1)\theta = 2\cos(2n-1)\theta \sin \theta \quad \textbf{[DO NOT PROVE THIS]}$$

to show that $u_n - u_{n-1} = (-1)^{n-1} \frac{2}{2n-1}$, $n = 2, 3, 4, \dots$ 2

(ii) Deduce that $u_n = 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{(-1)^{n-1}}{2n-1} \right)$ 3

End of Question 7

(a) A recurrence relationship is given by

$$u_{n+1} = \frac{u_n}{2} + \frac{1}{u_n}, \quad n = 1, 2, 3, \dots \quad \text{where } u_1 = 1$$

(i) Find u_3 **1**

(ii) It can be shown that $u_n = \sqrt{2} \left(\frac{1 + A}{1 - A} \right)$

$$\text{where } A = (-1)^{2^{n-1}} (\sqrt{2} - 1)^{2^n}$$

[DO NOT PROVE THIS]

Show that $u_{n+1} = \sqrt{2} \left(\frac{1 + A^2}{1 - A^2} \right)$ **1**

(iii) Use mathematical induction to prove that

$$u_n = \sqrt{2} \frac{(1 + (-1)^{2^{n-1}} (\sqrt{2} - 1)^{2^n})}{1 - (-1)^{2^{n-1}} (\sqrt{2} - 1)^{2^n}}, \quad n \geq 1$$
 4

(iv) Find $\lim_{n \rightarrow \infty} u_n$ **1**

Question 8 continues on the next page

(b) Let $f(\theta) = \frac{14 - 12\sin\theta - 6\cos\theta}{9 - 8\sin\theta - 3\cos\theta}$

(i) Use the subsidiary angle method to show that

$$9 - 8\sin\theta - 3\cos\theta > 0 \text{ for all } \theta \quad 2$$

(ii) Alternative expressions for $f(\theta)$ are

$$1 + \frac{5 - 4\sin\theta - 3\cos\theta}{9 - 8\sin\theta - 3\cos\theta} \text{ and } 2 - \frac{4 - 4\sin\theta}{9 - 8\sin\theta - 3\cos\theta}$$

[DO NOT VERIFY THESE] 2

Deduce that $1 \leq f(\theta) \leq 2$ for all θ

(iii) Verify that $f(\theta) = 1$ when $\sin\theta = \frac{4}{5}$, $0 < \theta < \frac{\pi}{2}$ 1

(iv) Sketch the graph of $y = f(\theta)$, $-\pi \leq \theta \leq \pi$, clearly indicating the y intercept. 3

End of Examination Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



THE KING'S SCHOOL

2008 Higher School Certificate Trial Examination

Mathematics Extension 2

Question	(Marks)	Complex Numbers	Functions	Integration	Conics	Mechanics
1	(15)		(b)(i), (c)(i) 3	(a), (b)(ii), (c)(ii), (d) (e) 12		
2	(15)	(a), (b), (c) 12	(d) 3			
3	(15)		(a), (c) 10	(b) 5		
4	(15)		(a)(i), (iv) 2	(a)(ii), (iii) 4		(b) 9
5	(15)			(a) 6		(b) 9
6	(15)				(a), (b) 15	
7	(15)	(a) 10	(b)(ii) 3	(b)(i) 2		
8	(15)		(a), (b) 15			
Total	(120)	22	36	29	15	18

TKS EXTENSION 2 SOLUTIONS 2009

Question 1

$$\begin{aligned} (a) \quad I &= \int \frac{x+1-1}{(x+1)^2} dx = \int \frac{1}{x+1} - (x+1)^{-2} dx \\ &= \ln(x+1) + \frac{1}{x+1} + c \end{aligned}$$

$$(b) \quad (i) \quad \text{Put } \frac{2x+9}{(2x-1)(x+2)} \equiv \frac{A}{2x-1} + \frac{B}{x+2}$$

$$: A(x+2) + B(2x-1) \equiv 2x+9$$

$$x = -2 \Rightarrow -5B = 5, \quad B = -1 \quad \therefore A - 2 = 2, \quad A = 4$$

$$\therefore \frac{4}{2x-1} - \frac{1}{x+2}$$

$$(ii) \quad \text{From (i)}, \quad I = 2 \ln(2x-1) - \ln(x+2) + c$$

$$\begin{aligned} (c) \quad (i) \quad \cos^3 x \sin^{12} x &= \cos x (1 - \sin^2 x) \sin^{12} x \\ &= \cos x \sin^{12} x - \cos x \sin^{14} x \end{aligned}$$

$$(ii) \quad \text{From (i)}, \quad I = \left[\frac{\sin^{13} x}{13} - \frac{\sin^{15} x}{15} \right]_0^{\pi/2}$$

$$= \frac{1}{13} - \frac{1}{15} - (0) = \frac{2}{195}$$

$$\begin{aligned} (d) \quad x &= u^2 & x=1, u=1 \\ \therefore \frac{dx}{du} &= 2u & x=3, u=\sqrt{3} \end{aligned}$$

$$\therefore I = \int_1^{\sqrt{3}} \frac{2u}{(u^2+1)u} du = 2 \left[\tan^{-1} u \right]_1^{\sqrt{3}}$$

$$= 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$$

$$= \frac{\pi}{6}$$

(e) Put $u = \ln x$, $\frac{dv}{dx} = x^{-2}$

$\therefore \frac{du}{dx} = \frac{1}{x}$, $v = -\frac{1}{x}$

$\therefore I = \left[-\frac{\ln x}{x} \right]_1^e + \int_1^e \frac{1}{x^2} dx$

$= -\frac{1}{e} - \left[\frac{1}{x} \right]_1^e = -\frac{1}{e} - \left(\frac{1}{e} - 1 \right) = 1 - \frac{2}{e}$

Question 2

(a) (i) $|z| = \sqrt{2+2} = 2$; $\arg z = \frac{\pi}{4}$

(ii) $z = 2 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$

$\therefore z^{12} = 2^{12} (\cos 3\pi + i \sin 3\pi) = -2^{12}$

(b) Put $a + ib = \sqrt{1 + 2\sqrt{2}i}$

Then $(a + ib)^2 = a^2 - b^2 + 2abi = 1 + 2\sqrt{2}i$

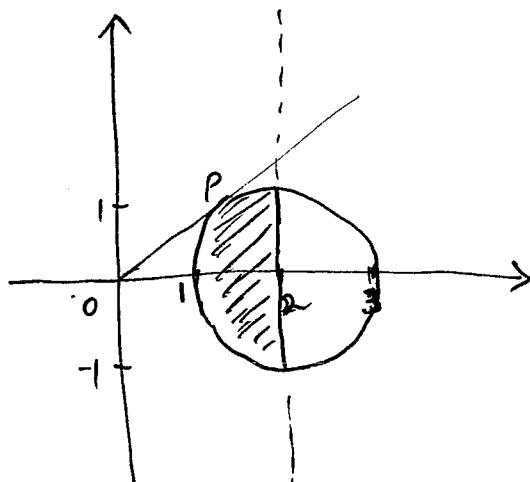
$\Rightarrow a^2 - b^2 = 1$

and $ab = \sqrt{2}$

$\therefore$ by inspection $a = \sqrt{2}$, $b = 1$

$\therefore \sqrt{1 + 2\sqrt{2}i} = \pm (\sqrt{2} + i)$

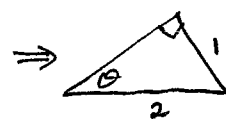
(c) (i)



(ii) Max $|z|$ occurs at $(2, 1)$
or $(2, -1)$

$\therefore \text{Max } |z| = \sqrt{2^2 + 1^2} = \sqrt{5}$

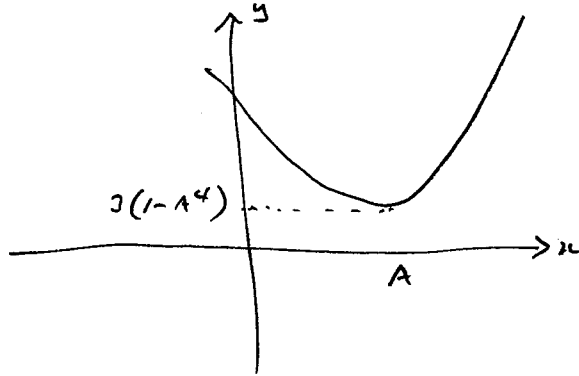
Max $\arg z$ occurs at P on diagram



$\Rightarrow \text{max } \arg z = \frac{\pi}{6}$

$$(d) \quad P'(x) = 4x^3 - 4A^3 \\ = 4(x^3 - A^3) = 0 \quad \text{if } x = A, \quad P(A) = A^4 - 4A^4 + 3 \\ = 3 - 3A^4$$

$\Rightarrow$ for 4 complex roots of $P(x) = 0$ we must have



$$\text{so, } 1 - A^4 > 0 \quad \text{or } A^4 < 1$$

$$\Rightarrow -1 < A < 1$$

Question 3

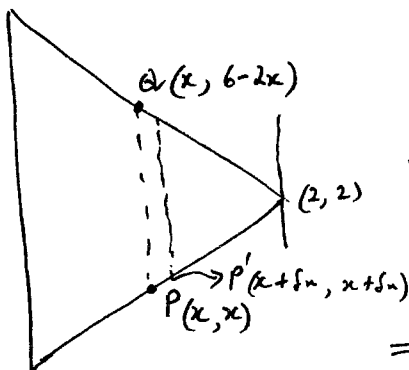
$$(a) \quad \text{Put } x = \frac{1}{1-x} \quad \therefore 1-x = \frac{1}{x} \quad \text{or } x = \frac{x-1}{x}$$

$$\therefore \text{cubic is } \left(\frac{x-1}{x}\right)^3 + \frac{x-1}{x} + 1 = 0$$

$$\text{or } x^3 - 3x^2 + 3x - 1 + x^3 - x^2 + x^3 = 0$$

$$\text{i.e. } 3x^3 - 4x^2 + 3x - 1 = 0$$

(b)



$$PQ = 6 - 2x - x = 3(2-x)$$

$$\therefore \delta V \approx \pi ((2-x)^2 - (2-x-\delta x)^2) 3(2-x)$$

$$\approx \pi (2(2-x)\delta x) 3(2-x)$$

$$\Rightarrow V = 6\pi \int_0^2 (x-2)^2 dx \quad \text{for ease}$$

$$= \frac{6\pi}{3} [(x-2)^3]_0^2 = 16\pi$$

(C) (i) For $x^2 - y^2 = a^2$

$$2x - 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y} = \frac{x_1}{y_1} \text{ at } P$$

$$\text{Gradient } OP = \frac{y_1}{x_1}$$

$\therefore$ product of gradients is 1

(ii) Now $\tan \alpha = \frac{x_1}{y_1}$ and $\tan \beta = \frac{y_1}{x_1}$

$$\therefore \tan \alpha = \cot \beta = \tan\left(\frac{\pi}{2} - \beta\right)$$

$$\Rightarrow \alpha = \frac{\pi}{2} - \beta \text{ or } \alpha + \beta = \frac{\pi}{2}$$

[LOTS OF ALTERNATIVES]

Question 4

(a) (i) G.S, $r = -x$, $N = 2n+1$

$$\therefore 1 - x + \dots + x^{2n} = \frac{1 - (-x)^{2n+1}}{1 - -x} = \frac{1 + x^{2n+1}}{1+x} \text{ since } 2n+1 \text{ is odd}$$

$$\begin{aligned} \text{(ii) From (i), } J &= \int_0^1 1 - x + x^2 - \dots + x^{2n} - \frac{1}{1+x} dx \\ &= \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + \frac{x^{2n+1}}{2n+1} - \ln(1+x) \right]_0^1 \\ &= 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n+1} - \ln 2 \end{aligned}$$

(iii) Since $0 \leq x \leq 1$, $\frac{x^{2n+1}}{1+x} \geq 0 \therefore J > 0$

$$\text{Also } \frac{x^{2n+1}}{1+x} < x^{2n+1}$$

$$\therefore J < \int_0^1 x^{2n+1} dx = \left[\frac{x^{2n+2}}{2n+2} \right]_0^1 = \frac{1}{2n+2}$$

$$\therefore 0 < J < \frac{1}{2n+2}$$

(iv) From (ii) and (iii)

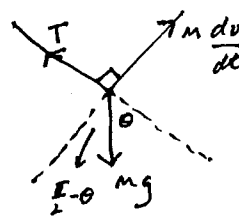
$$0 < 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n+1} - \ln 2 < \frac{1}{2n+2}$$

$$\therefore \text{since } \lim_{n \rightarrow \infty} \frac{1}{2n+2} = 0,$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n+1} - \ln 2 \right) = 0$$

$$\Rightarrow \ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \dots + \dots \text{ since } \lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0$$

(b) (i) $x = r\theta \quad \therefore \frac{dx}{dt} = r \frac{d\theta}{dt} \quad \text{i.e. } v = r \frac{d\theta}{dt}$

(ii)  $\therefore m \frac{dv}{dt} = -mg \cos\left(\frac{\pi}{2} - \theta\right)$
 $\text{i.e. } \frac{dv}{dt} = -g \sin \theta$

(iii) From (i), $\frac{dv}{dt} = r \frac{d^2\theta}{dt^2} = -g \sin \theta$

$$\therefore \frac{d^2\theta}{dt^2} = -\frac{g}{r} \sin \theta$$

(iv) If θ is small, $\sin \theta \approx \theta$

$$\therefore \frac{d^2\theta}{dt^2} \approx -\frac{g}{r} \theta \text{ is of the form } -n^2 \theta$$

$$\Rightarrow \text{SHM where } n = \sqrt{\frac{g}{r}}$$

$$\therefore \text{period} = 2\pi \sqrt{\frac{r}{g}}$$

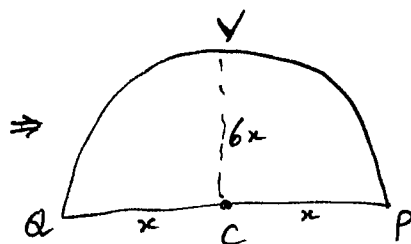
(v) Resolving in direction BC

$$m \frac{v^2}{r} = T - mg \cos \theta \approx T - mg \text{ for small } \theta$$

$$\therefore T \approx m \left(g + \frac{v^2}{r} \right)$$

Question 5

(a) (i)



$\therefore$ Using Simpson's rule,

$$\text{Area} = \frac{1}{6} \cdot 2x (0 + 0 + 24x) \\ = 8x^2$$

$$(ii) \therefore \delta V \approx 8x^2 \delta y = 32y \delta y$$

$$\therefore V = 32 \int_0^1 y \, dy = 16 [y^2]_0^1 = 16$$

$$(b) (i) m\ddot{x} = mg - mkv \Rightarrow \ddot{x} = g - kv$$

$$(ii) \ddot{x} = 0 \Rightarrow g - kV = 0 \quad \text{ie } V = \frac{g}{k}$$

$$(iii) \ddot{x} = \frac{dv}{dt} = k \left(\frac{g}{k} - v \right) = k(V - v)$$

$$\therefore k \frac{dt}{dv} = \frac{1}{V - v}$$

$$\Rightarrow k[t]_0^t = -[\ln(V - v)]_0^v$$

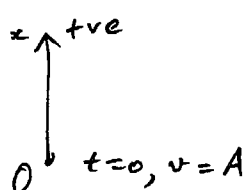
$$\therefore kt = -(\ln(V - v) - \ln V)$$

$$\text{or } \ln\left(\frac{V - v}{V}\right) = -kt$$

$$\therefore \frac{V - v}{V} = e^{-kt}$$

$$\Rightarrow v = V(1 - e^{-kt})$$

(iv)



$$m\ddot{x} = -mg - mkv$$

$$\therefore \ddot{x} = -g - kv = -k(V + v)$$

$$\therefore -k \frac{dt}{dv} = \frac{1}{V+v}$$

$$\Rightarrow -k [t]_0^T = [\ln(V+v)]_A^0$$

$$\Rightarrow -kT = \ln V - \ln(V+A) = \ln\left(\frac{V}{V+A}\right)$$

$\therefore$ From (iv), $\ln\left(\frac{V-v}{V}\right) = \ln\left(\frac{V}{V+A}\right)$ where v is the velocity of the first particle

$$\Rightarrow \frac{V-v}{V} = \frac{V}{V+A}$$

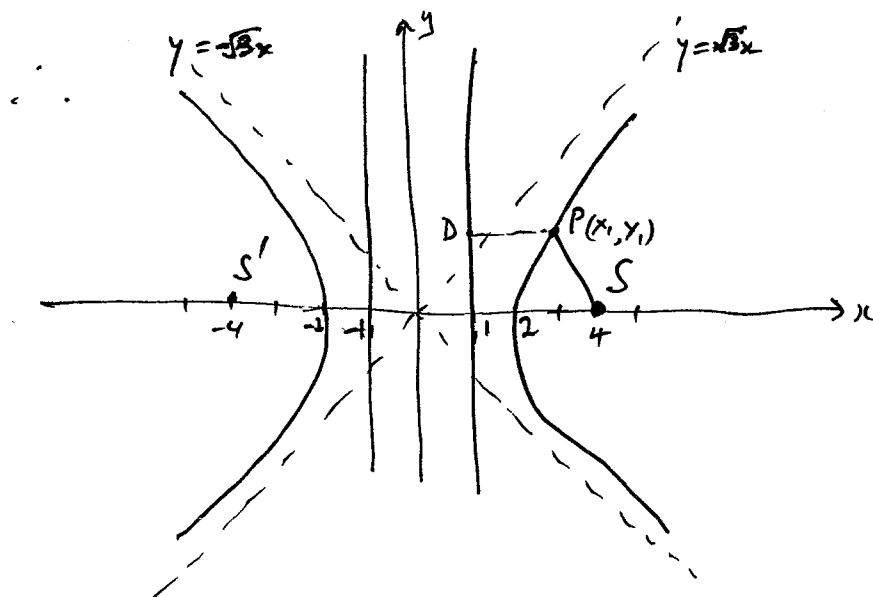
$$\therefore \frac{v}{V} = 1 - \frac{V}{V+A} = \frac{A}{V+A}$$

$$\therefore v = \frac{AV}{A+V}$$

Question 6

(a) (i) $a = 2$, $b^2 = 12 \Rightarrow c^2 = 4 + 12 = 16$, $c = 4$; $e = \frac{c}{a} = 2$

$\therefore$ Foci $(\pm 4, 0)$, directrices $x = \pm \frac{2}{2} = \pm 1$, asymptotes $y = \pm \frac{\sqrt{12}}{2}x = \pm \sqrt{3}x$



(ii) $PS = e PD = 2(x_1 - 1)$

$$(b) (i) \quad \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$= -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} \text{ at } P$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore \text{Tangent at } P \text{ is } y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\therefore ay \sin \theta - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

$$\Rightarrow bx \cos \theta + ay \sin \theta = ab (\cos^2 \theta + \sin^2 \theta) = ab$$

$$\text{i.e. } bx \cos \theta + ay \sin \theta - ab = 0$$

$$(ii) \quad pq = \frac{(abe \cos \theta - ab)(-abe \cos \theta - ab)}{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$= \frac{a^2 b^2 (1 - e \cos \theta)(1 + e \cos \theta)}{b^2 \cos^2 \theta + a^2 (1 - \cos^2 \theta)}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{a^2 - (a^2 - b^2) \cos^2 \theta}$$

$$= \frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{a^2 - a^2 e^2 \cos^2 \theta} = b^2$$

$$(iii) \quad \text{For } P(a, 0), \quad pq = (a - ae)(a + ae) \\ = a^2 - a^2 e^2 \\ = b^2$$

$$(iv) \quad (p - q)^2 = p^2 + q^2 - 2pq \\ = b(a^2 e^2) - 2(a^2 - a^2 e^2) \text{ from (iii)} \\ = 2a^2 (3e^2 - 1 + e^2) = 2a^2 (4e^2 - 1)$$

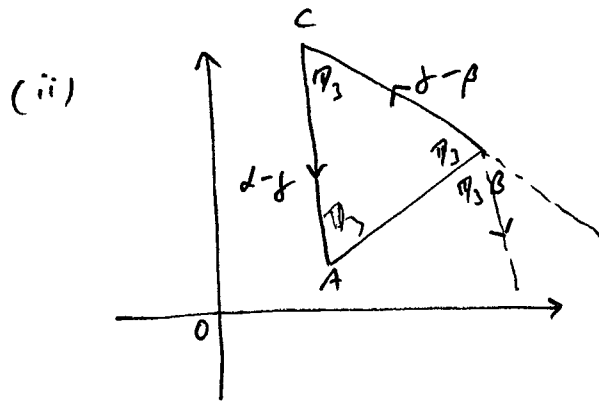
$$\text{But } (p - q)^2 \geq 0 \Rightarrow 4e^2 - 1 \geq 0 \quad \vee \quad e^2 \geq \frac{1}{4} \quad \text{i.e. } e \geq \frac{1}{2}$$

Question 7

(a) (i) $\omega^3 = \cos 2\pi + i \sin 2\pi = 1$

$$1 + \omega + \omega^2 = 1 + \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} + \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$$

$$= 1 - \frac{1}{2} + i \frac{\sqrt{3}}{2} - \frac{1}{2} - i \frac{\sqrt{3}}{2} = 0 \quad \left(\begin{array}{l} \text{ALTERNATIVES,} \\ \text{OF COURSE} \end{array} \right)$$



Now $\vec{CA} = \omega \vec{BC}$

$$\Rightarrow \alpha - \beta = \omega (\beta - \alpha)$$

(iii) From (ii) $\alpha - \beta = \omega \beta - \omega \alpha$

$$\therefore \alpha + \omega \beta - \beta(1 + \omega) = 0$$

$$\Rightarrow \alpha + \omega \beta - \beta(-\omega^2) = 0 \quad \text{from (i)}$$

$$\text{i.e. } \alpha + \omega \beta + \omega^2 \beta = 0$$

(iv) $\therefore \alpha + \omega \beta + \omega^2 \beta = 0$ and is the sum of the roots of $z^3 + pz + q = 0$

(v) $\alpha \cdot \omega \beta \cdot \omega^2 \beta = \alpha \beta^2 \omega^3 = -q \Rightarrow \alpha \beta^2 = -q$ from (i)

$$\text{i.e. } q = -\alpha \beta^2$$

(vi) Now $\sum \alpha^3 + p \sum \alpha + 3q = 0$

$$\Rightarrow \alpha^3 + \omega^3 \beta^3 + \omega^6 \beta^3 + p(0) - 3\alpha \beta^2 = 0$$

$$\text{i.e. } \alpha^3 + \beta^3 + \beta^3 = 3\alpha \beta^2 \quad \text{since } \omega^3 = 1$$

$$(ii) \sum_{n=2}^n (u_n - u_{n-1}) = u_n - u_1 = \sum_{n=2}^n (-1)^{n-1} \frac{2}{2n-1}$$

$$\therefore u_n = u_1 + \sum_{n=2}^n (-1)^{n-1} \frac{2}{2n-1}$$

$$= \int_0^{\pi/2} \frac{\sin 2\theta}{\sin \theta} d\theta + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{2}{2n-1}$$

$$= \int_0^{\pi} 2 \cos \theta \, d\theta + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{2}{2n-1}$$

$$= 2 \left[\sin \theta \right]_0^{\pi/2} + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{2}{2n-1}$$

$$= 2 + 2 \left(-\frac{1}{3} + \frac{1}{5} - \dots + (-1)^{n-1} \frac{1}{2n-1} \right)$$

$$= 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + (-1)^{n-1} \frac{1}{2n-1} \right)$$

Question 8

$$(a) \quad (i) \quad u_2 = \frac{1}{2} + 1 = \frac{3}{2}$$

$$\therefore u_3 = \frac{3}{4} + \frac{2}{3} = \frac{17}{12}$$

$$(ii) \quad u_{n+1} = \frac{1}{\sqrt{2}} \left(\frac{1+A}{1-A} \right) + \frac{1}{\sqrt{2}} \left(\frac{1-A}{1+A} \right)$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{(1+A)^2 + (1-A)^2}{1-A^2}$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{2(1+A^2)}{1-A^2} = \sqrt{2} \left(\frac{1+A^2}{1-A^2} \right)$$

$$(iii) \quad u_1 = \sqrt{2} \frac{1 - (\sqrt{2}-1)^2}{1 + (\sqrt{2}-1)^2} = \sqrt{2} \frac{2\sqrt{2}-2}{4-2\sqrt{2}} = \frac{4-2\sqrt{2}}{4-2\sqrt{2}} = 1$$

$$\therefore \text{Assume } u_n = \sqrt{2} \left(\frac{1+A}{1-A} \right) \text{ where } A = (-1)^{2^{n-1}} (\sqrt{2}-1)^{2^n} \text{ for integers } n \geq 1$$

$$\text{Then } u_{n+1} = \sqrt{2} \left(\frac{1+A^2}{1-A^2} \right) \text{ from (ii) \& using the assumption}$$

$$= \sqrt{2} \frac{1 + (-1)^{2^n} (\sqrt{2}-1)^{2^{n+1}}}{1 - (-1)^{2^n} (\sqrt{2}-1)^{2^{n+1}}}$$

$$\begin{aligned} \text{since } A^2 &= \left((-1)^{2^{n-1}} (\sqrt{2}-1)^{2^n} \right)^2 \\ &= (-1)^{2^n} \cdot 2 (\sqrt{2}-1)^{2^{n+1}} \\ &= (-1)^{2^n} (\sqrt{2}-1)^{2^{n+1}} \end{aligned}$$

$\therefore$ by induction it's correct.

$$(iv) \quad 0 < \sqrt{2}-1 < 1 \text{ and } 2^n \rightarrow \infty \text{ as } n \rightarrow \infty$$

$$\therefore \lim_{n \rightarrow \infty} (\sqrt{2}-1)^{2^n} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = \sqrt{2} \left(\frac{1}{1} \right) = \sqrt{2}$$

$$(b) (i) \quad 9 - 8 \sin \theta - 3 \cos \theta = 9 - \sqrt{8^2 + 3^2} \sin(\theta + \alpha) \text{ for some } 0 < \alpha < \frac{\pi}{2}$$

$$\geq 9 - \sqrt{73} \text{ since } |\sin(\theta + \alpha)| \leq 1$$

$$> 0 \quad \forall \theta$$

$$(ii) \quad f(\theta) = 1 + \frac{5 - 5 \sin(\theta + \alpha)}{9 - 8 \sin \theta - 3 \cos \theta} \geq 1 \text{ since } |\sin(\theta + \alpha)| \leq 1 \quad \& (i)$$

$$f(\theta) = 2 - \frac{4 - 4 \sin \theta}{9 - 8 \sin \theta - 3 \cos \theta} \leq 2 \text{ since } |\sin \theta| \leq 1 \quad \& (i)$$

$$\text{i.e.} \quad 1 \leq f(\theta) \leq 2 \quad \forall \theta$$

$$(iii) \quad \text{If } \sin \theta = \frac{4}{5} \text{ then } \cos \theta = \frac{3}{5} \text{ since } 0 < \theta < \frac{\pi}{2}$$

$$\text{Then } f(\theta) = 1 + \frac{5 - \frac{16}{5} - \frac{9}{5}}{9 - 8 \cdot \frac{4}{5} - \frac{9}{5}} = 1$$

$$(iv) \quad f(\pi) = f(-\pi) = \frac{14 + 6}{9 + 3} = \frac{5}{3}$$

$$f(0) = \frac{14 - 6}{9 - 3} = \frac{4}{3}$$

$$\sin \theta = \frac{4}{5} \Rightarrow \theta \approx .93$$

$$f(\theta) = 2 \text{ if } \sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2}$$

